

From Search Goals to Generative Intent: Proposing the Generative Intent Operationalization (GIO) A Comprehensive Analysis of Epistemic Necessity as the Core Driver for Generative Engine Optimization (GEO)

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Abstract

The paradigm shift from classical Information Retrieval (IR) toward Generative Engines (GE) necessitates a fundamental re-evaluation of user intent. While classical paradigms viewed the search engine as a deterministic router, Generative Engines act as probabilistic answer machines. This shift renders established taxonomies, such as Broder's "Navigational/Informational/Transactional" model, insufficient for predicting the behavior of Large Language Models (LLMs). This position paper proposes the Generative Intent Operationalization (GIO), a predictive, pre-generation taxonomic framework that classifies user intent before the model produces output-in contrast to the majority of existing approaches that analyze logs or interaction traces post hoc. Unlike recent rival taxonomies that classify based on sociological interaction (TUNA), economic tasks (NBER), or dynamically generated categories (Shah et al.), GIO classifies intent based on Grounding Necessity (GN): the epistemic requirement for a model to retrieve external evidence to validate a response. By synthesizing recent findings in agentic architectures-specifically the "Epistemic Effort Invariant" (Wang et al., 2025), "Self-RAG" mechanisms (Asai et al., 2024), and confidenceaware retrieval (FLARE, Jiang et al., 2023)-we demonstrate that GN is a critical constraint governing visibility in AI-generated answers. This predictive orientation makes GIO directly actionable for Generative Engine Optimization (GEO) tooling: rather than retroactively explaining which content appeared in generative answers, GIO enables strategists to anticipate retrieval demand and optimize proactively. We present GIO as the theoretical superstructure for GEO, distinguishing between parametric memorization and high-value grounded generation.

Keywords

computing and processing, epistemic uncertainty, generative engine optimization, grounding necessity, large language models, rag, search intent

From Search Goals to Generative Intent: Proposing the Generative Intent Operationalization (GIO)

A Comprehensive Analysis of Epistemic Necessity as the Core Driver for Generative Engine Optimization (GEO)

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Abstract

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This position paper proposes the **Generative Intent Operationalization (GIO)**, a *predictive, pre-generation* taxonomic framework that classifies user intent *before* the model produces output—in contrast to the majority of existing approaches that analyze logs or interaction traces *post hoc*. Unlike recent rival taxonomies that classify based on sociological interaction (TUNA), economic tasks (NBER), or dynamically generated categories (Shah et al.), GIO classifies intent based on **Grounding Necessity (GN)**: the epistemic requirement for a model to retrieve external evidence to validate a response. By synthesizing recent findings in agentic architectures—specifically the “Epistemic Effort Invariant” (Wang et al., 2025), “Self-RAG” mechanisms (Asai et al., 2024), and confidence-aware retrieval (FLARE, Jiang et al., 2023)—we demonstrate that GN is a critical constraint governing visibility in AI-generated answers. This predictive orientation makes GIO directly actionable for *Generative Engine Optimization (GEO)* tooling: rather than retroactively explaining which content appeared in generative answers, GIO enables strategists to anticipate retrieval demand and optimize proactively. We present GIO as the theoretical superstructure for GEO, distinguishing between parametric memorization and high-value grounded generation.

CCS Concepts

• **Information systems** → **Information retrieval**; • **Human-centered computing** → *HCI theory, concepts and models*.

Keywords

Generative Engine Optimization, Grounding Necessity, Epistemic Uncertainty, Search Intent, RAG, Large Language Models

1 Introduction: The Epistemic Shift

The transformation of Information Retrieval (IR) systems into Generative Engines (GE) marks a significant structural shift in human-machine interaction. While the classical IR paradigm [3, 10] understood the system as a router directing users to external locations (URLs), GEs function as probabilistic answer machines—a transformation anticipated by Metzler et al.’s vision of search systems that produce expert-level answers rather than ranked document lists [17].

This shift fundamentally alters the nature of “User Intent.” The user no longer primarily seeks a location, but a synthesis. For the content strategist, this creates a critical opacity: the “Black Box” of generation. Does the answer stem from the model’s parametric memory (*Memorization*), or was it constructed through external retrieval (*Grounding*)?

Current taxonomies fail to address this distinction. This paper identifies the central deficit: We lack a framework to predict *Grounding Necessity* based on the prompt alone. We propose the **Generative Intent Operationalization (GIO)**,¹ a taxonomic framework aligning user goals with the engine’s epistemic requirements. GIO serves as the theoretical superstructure for the emerging discipline of Generative Engine Optimization (GEO).

To clarify the contribution of this work, we explicitly delimit the scope of GIO:

- **Not a System Architecture:** GIO does not propose a new retrieval algorithm or vector database structure.
- **Not a Post-Hoc Analysis:** Unlike studies that analyze logs of completed conversations, GIO models intent *a priori*—before the model generates a response. This pre-generation orientation is a central differentiator: it transforms GIO from an explanatory taxonomy into a predictive tool for content strategy.
- **Not a Prompt Engineering Guide:** GIO focuses on the informational nature of the request, not syntactical optimization.

Instead, GIO contributes the first *pre-generation operationalization* that classifies user intent based on probabilistic grounding necessity—independent of the specific model architecture used. An important boundary condition applies: GIO targets the *public retrieval* path. As LLMs increasingly access private user data (emails,

¹The term “Generative Intent” also appears in Kou et al.’s “GIA” (IEEE, 2025) [16], which addresses network slicing—an entirely unrelated usage.

calendars), a “Personalization Wall” emerges where GN is high but GEO opportunity is zero. We return to this limitation in Section 7.1.

To illustrate GIO’s practical value, we use a running example throughout this paper: the query “Best SEO strategy for banks in 2026.” Under Broder’s taxonomy [3], this is classified as “Informational”—a label that tells a content strategist nothing actionable. GIO classifies it as **Mode 1.3 Advisory** with **High Grounding Necessity**: the model cannot answer from parametric memory alone, because SEO best practices are volatile ($V_{volatility}$: high), domain-specific (E_{spec} : high), and the prompt implicitly demands current expert consensus (I_{gap} : high). The strategic consequence—producing authoritative, citable thought leadership content—follows directly from the classification.

2 The Foundation: Generative Engine Optimization (GEO)

To scientifically locate GIO, we must first define the field. The literature on GEO has grown rapidly between 2024 and 2026, distinguishing itself clearly from traditional SEO.

2.1 The Foundational Empirical Work: Aggarwal et al.

Aggarwal et al. (2024) [1] provide the empirical proof that visibility in generative answers can be manipulated. Introducing *GEO-bench*, they demonstrated that tactics such as adding citations and statistical data can increase visibility by up to 40%. However, Aggarwal focuses on *tactics* (what to change). GIO focuses on *strategy* (when to optimize). Without GIO, practitioners apply Aggarwal’s tactics blindly to queries with low grounding necessity, resulting in wasted effort.

2.2 GEO vs. AEO: A Critical Distinction

It is crucial to distinguish GEO from Answer Engine Optimization (AEO).

- **AEO (Extraction):** Targets “Featured Snippets” (Position 0). It relies on the literal extraction of a sentence to answer a Fact Retrieval query (GIO Mode 1.1).
- **GEO (Synthesis):** Targets the inclusion of a document as evidence in a newly generated argument. As shown by Bagga et al. in “E-GEO” [2], this becomes relevant in complex, transactional scenarios where the engine must synthesize dynamic attributes (e.g., price, availability).

Returning to our running example: the query “Best SEO strategy for banks in 2026” falls squarely into GEO territory. An AEO approach would optimize for a single extractable sentence (“The best strategy is...”). A GEO approach, informed by GIO’s classification as Mode 1.3 Advisory, would instead optimize for inclusion as a cited source in a multi-perspective synthesis—requiring the content to demonstrate expert authority, provide current data, and structure arguments for machine readability.

3 The Competition: Alternative Taxonomies

The year 2025 produced significant new classification models for LLM interactions. GIO competes directly with these frameworks for interpretive authority.

3.1 The Sociological Rival: TUNA

The *Taxonomy of User Needs and Actions (TUNA)* by Shelby, Diaz, and Prabhakaran (2025) [5] proposes 6 modes and 57 query types, including “Social Interaction” and “Meta-Conversation.” **Critique:** While TUNA is descriptively excellent for HCI research, it is operationally too granular for optimization. A GEO strategist cannot optimize for “Social Banter.” GIO collapses TUNA’s social categories into a single dimension: “Low Grounding Necessity,” rendering them non-actionable for optimization purposes. TUNA and GIO are thus complementary: TUNA describes the full breadth of human-AI interaction, while GIO filters for the subset that is strategically relevant to content visibility.

3.2 The Economic Rival: NBER

A large-scale empirical study by Chatterji et al. (NBER, 2025) [4] classifies intent by economic utility, identifying “Writing” and “Seeking Information” as dominant tasks. **Critique:** The category “Writing” is dangerously ambiguous. It conflates *Ungrounded Generation* (e.g., “Write a poem”) with *Grounded Generation* (e.g., “Summarize this report”). GIO splits this category, as only the latter offers GEO potential.

3.3 The Dynamic Rival: Automated Taxonomies

Shah et al. (2025) [6] argue against static taxonomies entirely, proposing LLM-generated dynamic taxonomies that adapt to evolving usage patterns. **Defense of GIO:** While topics change dynamically, the epistemic nature of a query remains an invariant property. A query for “Current CEO of X” will *always* require grounding, regardless of the year. GIO is thus a robust *meta-framework* that operates at the epistemic level rather than the topical level, making it complementary to—not replaced by—dynamic classification.

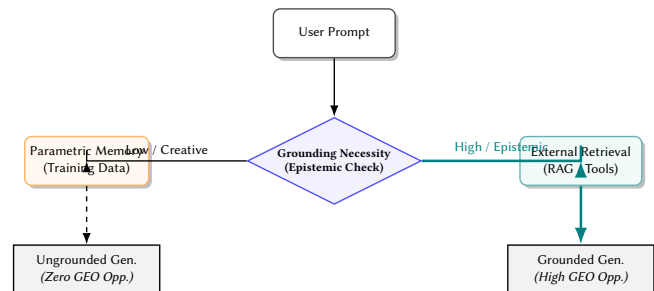


Figure 1: The GIO Decision Flow. High Grounding Necessity triggers the retrieval path, creating the GEO opportunity.

4 The Architectural Basis of Grounding

GIO postulates “Grounding Necessity” (GN) as an epistemic property of the prompt. This is *corroborated* by recent computer science literature which treats GN as a hard architectural constraint in the generation pipeline.

4.1 The Epistemic Effort Invariant (ToA)

Wang et al. (2025) in *Theory of Agent (ToA)* [7] introduce the “Epistemic Effort Invariant.” They argue that an agent should invoke

external tools *only* when internal reasoning is insufficient to resolve epistemic uncertainty. GIO describes the publisher’s strategy for the exact phenomenon that ToA describes as the agent’s policy. The convergence is notable: both frameworks independently arrive at epistemic necessity as the decision boundary—ToA from the system’s perspective, GIO from the content strategist’s perspective.

4.2 Operationalizing Uncertainty: Self-RAG and FLARE

Technical implementations confirm GIO’s relevance through two complementary mechanisms:

- **Self-RAG** (Asai et al., 2024) [8] trains models to generate “Reflection Tokens” (e.g., Retrieve=Yes). This is the algorithmic implementation of high GN: the model explicitly signals when parametric generation is insufficient.
- **FLARE** (Jiang et al., 2023) [9] triggers retrieval only when the model’s confidence in the next token drops below a threshold. This provides a continuous, token-level operationalization of grounding necessity.
- **Adaptive-RAG** (Jeong et al., 2024) [13] uses a lightweight classifier to route queries into “Simple” (parametrically solvable) and “Complex” (retrieval-required) categories. This maps directly onto the Intent-Complexity Nexus (Table 2): Level 1 queries correspond to Adaptive-RAG’s simple path (Low GN), while Level 3 queries trigger the multi-step retrieval path (High GN).

Together, these systems demonstrate that Grounding Necessity is not a marketing construct, but a measurable variable in the generation process—one that GIO attempts to predict *a priori* from the prompt alone. Early work on retrieval-augmented generation, notably WebGPT [18], already demonstrated that LLMs can learn to issue search queries when their parametric knowledge is insufficient—an implicit operationalization of GN at the system level. A caveat applies: Self-RAG’s reflection tokens are themselves learned predictions and may produce false positives (triggering retrieval for queries answerable from parametric memory) or false negatives (failing to trigger for genuinely novel queries). We address this limitation in our validation design (Section 6.3).

4.3 Retrieval Quality as a Constraint: Corrective RAG

Even when Grounding Necessity is correctly identified as high, the quality of retrieved evidence determines whether the generation succeeds or degrades into hallucination. Yan et al. (2024) [14] address this through *Corrective RAG* (CRAG), which introduces a retrieval evaluator that assesses the relevance of retrieved documents before they enter the generation pipeline. When the retrieved material is judged irrelevant, a fallback mechanism (e.g., web search) is triggered. CRAG thus validates a critical implication of GIO: identifying High GN is necessary but not sufficient. The “Zone of Grounding” (Table 2) requires not merely retrieval, but retrieval of authoritative, well-structured content—precisely the output that GEO aims to produce.

5 The Generative Intent Operationalization (GIO)

GIO classifies prompts based on the probability that a valid answer requires external data access (see Figure 1).

5.1 The Intent-Complexity Nexus

While GIO provides a taxonomic classification of goals, it must be mapped against **Structural Prompt Complexity**. A simple query often yields a parametric answer, while a complex Chain-of-Thought (CoT) prompt forces the model to verify facts. By intersecting Intent (GIO) and Complexity, we identify “Zones of Opportunity” (Table 2).

The matrix reveals that Grounding Necessity is not static. It increases with prompt complexity across all modes. Notably, the matrix also exposes the strategic implication: optimizing for keywords alone is insufficient; GEO requires optimizing for questions that necessitate external validation (Level 3 complexity).

5.2 Operationalizing GEO Strategies: A Worked Example

To illustrate how GIO classifications translate into concrete content decisions, we trace a second example through the framework. Consider a health insurance company whose content appears in generative answers to the query: “What does the new German digital health law mean for private insurers?”

GIO Classification: Mode 1.3 Advisory (expert perspective sought) at Level 3 Complexity (requires reasoning across legal, technical, and economic dimensions). I_{gap} : high (implicit evidential demand). $V_{volatility}$: high (recent legislation). E_{spec} : high (specific legal instrument). T_{decay} : high (post-cutoff event). Result: *Zone of Synthesis*, maximal GEO value.

Derived GEO Strategy (“Thought Leadership”): The strategy label from Table 1 decomposes into actionable requirements: (a) *Structural*: Organize content with machine-readable section headings that mirror likely sub-questions (“Who is affected?”, “What changes by when?”, “What are the compliance requirements?”), enabling the generative engine to extract modular answer components. (b) *Evidential*: Embed primary sources (legislative references, regulatory body statements) as inline citations, since the engine’s grounding mechanism preferentially selects content that itself cites authoritative origins. (c) *Perspectival*: Present multiple expert viewpoints (legal analysis, industry association position, patient advocacy perspective), matching the Advisory mode’s constraint of “Need for Diversity of Perspectives.” This increases the probability that the content serves as a cited source in a multi-perspective synthesis rather than being bypassed for a source that covers only one angle. (d) *Temporal*: Include explicit publication and last-updated dates, and mark time-sensitive claims (“As of March 2026, the implementation deadline is...”), reducing the engine’s uncertainty about content freshness.

This decomposition demonstrates that GIO’s classification is not merely descriptive but prescriptive: each parameter of the GN assessment (I_{gap} , $V_{volatility}$, E_{spec} , T_{decay}) maps to a specific content optimization lever.

Table 1: The Generative Intent Operationalization (GIO) – Classification Matrix

Mode	Category	User Intent & Context	Grounding Necessity (GN) & GEO Strategy
1. ASKING	1.1 Fact Retrieval	Intent: Verify a static fact (“Height of Eiffel Tower”). <i>Constraint: High Consensus.</i>	Low. The Parametric Trap. <i>Strategy: Entity Authority & Brand Awareness (Zero-Click).</i>
	1.2 Real-Time Synthesis	Intent: Access dynamic data (“Stock price”, “Election results”). <i>Constraint: High Volatility.</i>	High. Requires structured data retrieval. <i>Strategy: Schema Markup & Data Feeds.</i>
	1.3 Advisory	Intent: Seek expert perspective (“SEO Strategy 2026”). <i>Constraint: Subjective Nuance; Need for Diversity of Perspectives.</i>	High. Requires expert consensus and multi-perspective synthesis. <i>Strategy: Thought Leadership.</i>
2. DOING	2.1 Utility	Intent: Transformation of existing input (“Translate to Spanish”).	None. Tool usage. <i>Strategy: Actions Schema & Plugin Discovery.</i>
	2.2 Ungrounded Gen.	Intent: Creative creation ex nihilo (“Write a poem”).	Low. Relies on internal creativity. <i>Strategy: Style Prompting.</i>
	2.3 Grounded Gen.	Intent: Source-dependent creation (“Summarize this PDF”). <i>Constraint: User-provided source material.</i> <i>Note: GN is maximal, but grounding is user-initiated (the user supplies the document), not system-initiated (the model retrieves from an index). GEO opportunity applies only to the latter; for user-provided sources, optimization targets in-document structure rather than discoverability.</i>	Maximal (user-provided). Must process specific content. <i>Strategy: In-Document Optimization (structure, headings, machine-readable formatting).</i>
3. ACTING	3.1 Transactional	Intent: Execute a purchase or booking (“Book a flight”).	High. API dependent. <i>Strategy: Agent Actions.</i>
	3.2 Open-Ended Investigation	Intent: Solve an underspecified problem requiring iterative exploration (“Plan a trip to Japan”). <i>Note: Framed as user goal, not system strategy.</i>	High. Iterative retrieval required. <i>Strategy: Deep Content & Hubs.</i>

Table 2: The Unified GEO Matrix – Intersection of Intent (GIO) and Complexity

Prompt Complexity → GIO Intent ↓	Level 1: Zero-Shot (Simple)	Level 2: Context-Rich (Guided)	Level 3: Reasoning (Complex)
Mode 1: ASKING (Information)	<i>Zone of Hallucination</i> Model answers from training data. GEO Value: None	<i>Zone of Refinement</i> Model adapts tone/style. GEO Value: Low	<i>Zone of Synthesis</i> Must compare external sources. GEO Value: High
Mode 2: DOING (Generation)	<i>Zone of Creativity</i> Model generates ex nihilo. GEO Value: None	<i>Zone of Templating</i> Model follows constraints. GEO Value: Low	<i>Zone of Grounding</i> Processes specific sources. GEO Value: Maximal
Mode 3: ACTING (Execution)	<i>Zone of Delegation</i> Single API call. GEO Value: Low	<i>Zone of Orchestration</i> Multi-step with constraints. GEO Value: Medium	<i>Zone of Autonomy</i> Iterative retrieval and planning. GEO Value: High

5.3 The Parametric Trap

A central concept in GIO is the *Parametric Trap*: the condition in which a model generates a confident but epistemically invalid response—hallucinated, stale, or oversimplified—because the query

appears answerable from parametric memory. The trap is sprung when high entity specificity (E_{spec}) in the training corpus masks high volatility ($V_{volatility}$). For instance, “Who is the CEO of Google?” involves a head entity with abundant training data, leading models to answer confidently from parameters. Yet the answer is volatile:

CEO tenure is finite, and a parametric response may be outdated. The Parametric Trap thus describes a *false negative* in grounding necessity assessment—a query that *should* trigger retrieval but does not, precisely because the model’s confidence is miscalibrated by familiarity. For GEO, the implication is that content targeting Parametric Trap queries must establish *Entity Authority*—becoming the canonical source that the model’s training data (and future fine-tuning) absorbs as ground truth.

5.4 The Grounding Necessity Hypothesis

We reformulate the probability of Grounding Necessity (P_{GN}) as a *conceptual framework model*—a structured enumeration of the variables that govern retrieval necessity and their hypothesized interactions, rather than a calibrated predictive equation. A known limitation of naïve models is the “Specificity Paradox”: highly specific entities (e.g., “Paris”) often have low retrieval needs because they are head entities in the training data.

A crucial scope condition applies: GN targets *epistemic* uncertainty—uncertainty arising from incomplete knowledge that is resolvable through retrieval—not *aleatory* uncertainty, which reflects inherent randomness in the world (e.g., a coin flip). This distinction, well-established in Bayesian epistemology and increasingly operationalized in retrieval-augmented systems [15], ensures that GIO does not overpredict retrieval necessity for queries whose uncertainty is fundamentally irreducible.

We propose **Hypothesis H1**:

$$P_{GN} \approx f(I_{gap}) \cdot (\alpha \cdot T_{decay} + \beta \cdot (E_{spec} \times V_{volatility})) \quad (1)$$

The structure of Equation (1) is deliberately multiplicative: $f(I_{gap})$ acts as a gating factor on the entire expression. If the prompt signals no information gap ($I_{gap} \rightarrow 0$), then $f(I_{gap}) \rightarrow 0$ and $P_{GN} \approx 0$ —regardless of how volatile the topic is or how far beyond the training cutoff the referenced event lies. This captures a key intuition: temporal decay and entity volatility only drive retrieval when the prompt itself demands external evidence. A creative writing prompt about a volatile entity (“Write a sonnet about Bitcoin”) has high $V_{volatility}$ but near-zero I_{gap} , yielding low P_{GN} as expected. Where:

- I_{gap} : Information Gap—the explicit demand for data density (citations, tables, numerical evidence) in the prompt. Adapted from Loewenstein’s information gap theory of curiosity [11], which posits that curiosity arises from a perceived gap between what one knows and what one wants to know. In the LLM context, I_{gap} measures the degree to which the prompt signals a deficit that parametric memory cannot fill. While a full operationalization of I_{gap} is beyond the scope of this position paper, we note several candidate proxies amenable to algorithmic measurement: (a) the presence of evidential demand markers (“cite”, “according to”, “source”, “data”), (b) the ratio of interrogative constructions to imperative constructions (questions demand external facts; imperatives often request generation), and (c) the prompt’s specificity gradient—the degree to which the prompt constrains acceptable answers to a narrow factual range rather than permitting open-ended generation. We

acknowledge that these surface-level proxies may be insufficient for prompts that carry implicit information demands through conversational implicature—e.g., “What should I think about X?” implicitly requires evidence even without explicit evidential markers. Detecting such implicit I_{gap} likely requires semantic analysis beyond lexical heuristics, and represents an important avenue for future operationalization work.

- T_{decay} : Temporal distance from training cutoff to the event referenced in the prompt. A practical limitation applies: for proprietary models (e.g., GPT-4, Gemini), the exact training cutoff is often undisclosed or continuously shifting through incremental updates. In such cases, T_{decay} must be estimated via proxy—for example, by testing whether the model can answer dated factual queries (“Who won the 2025 Super Bowl?”) and inferring an approximate boundary from the pattern of correct vs. hallucinated responses.
- $E_{spec} \times V_{volatility}$: The interaction term. Entity specificity (E_{spec}) only drives GN when the entity is subject to change ($V_{volatility}$). We conceptualize $V_{volatility}$ as a continuous measure ranging from 0 (immutable facts, e.g., gravitational constant, historical dates) to 1 (real-time data, e.g., stock prices, live election results). Static, specific entities (e.g., historical facts about well-known places) remain in the Parametric Trap regardless of specificity.
- $f(\cdot)$: A monotonically increasing function mapping I_{gap} to $[0, 1]$. As the default testable instantiation, we propose a sigmoid function $f(I_{gap}) = \frac{1}{1+e^{-k(I_{gap}-\theta)}}$, which naturally bounds P_{GN} to the probability range $[0, 1]$ while capturing the intuition that GN increases sharply once the information gap crosses a threshold θ . We acknowledge that this specific functional form is a starting hypothesis; alternative forms (e.g., piecewise linear, logistic with interaction terms) may prove empirically superior. The critical theoretical commitment is the monotonicity and gating role of f , not its exact shape.
- α, β, k, θ : Parameters to be determined through empirical calibration against the ground truth described in Section 6.3. Until calibration data are available, the framework model serves to identify the *relevant dimensions* of GN and their structural relationships, rather than to produce numerical predictions.

6 Roadmap for Empirical Validation

The primary contribution of this position paper is the operationalization itself—the claim that Grounding Necessity is the correct organizing principle for classifying user intent in generative systems. The following section does not present results, but outlines a falsifiable validation framework. Its purpose is to demonstrate that GIO is not merely a conceptual exercise but a testable theory, and to provide a concrete protocol for future empirical work.

6.1 Sampling Strategy

We will sample $N = 1,000$ prompts from the **WildChat corpus** [12] using *stratified sampling* to ensure coverage of both head queries (short, common) and long-tail queries (long, specific). The sample

size exceeds the minimum of $N = 800$ determined by power analysis (see Section 6.3), providing a safety buffer for exclusions during annotation. A known limitation of WildChat is its distributional bias: coding and technical queries are overrepresented, while commercial and transactional queries are underrepresented [12]. To mitigate this, we will supplement the WildChat sample with queries drawn from e-commerce and transactional datasets (e.g., E-GEO [2]) to ensure adequate coverage of Mode 3 (ACTING) categories.

6.2 Phase 1: Inter-Rater Reliability

Three domain experts will annotate the sampled prompts using the GIO classification matrix (Table 1). We target a Fleiss’ Kappa $\kappa > 0.7$ to establish sufficient inter-rater reliability for a framework intended for operational use.

6.3 Phase 2: Ground Truth and Baseline Comparison

We use a dual ground truth approach:

- **Human Expert Judgment (Gold Standard):** Three domain experts independently annotate whether retrieval *should* have been triggered for each prompt, using the GIO classification matrix (Table 1) and their domain knowledge. Human judgment serves as the primary gold standard because it captures true epistemic necessity—whether a factually adequate response *requires* external evidence—rather than a system’s learned retrieval policy.
- **Technical Proxy (Scalability Baseline):** The **Self-RAG model** [8] generates reflection tokens (e.g., [Retrieve]) indicating the system’s internal assessment of retrieval necessity. This provides an automated, reproducible baseline that enables scaling beyond hand-annotated datasets. However, Self-RAG’s reflection tokens measure the *system’s calibration* rather than true epistemic necessity: a poorly calibrated model may over-trigger retrieval for queries it could resolve parametrically, or under-trigger for genuinely novel queries where its training data is sparse. This risk is especially acute for emerging topics and tail entities. We therefore treat Self-RAG as a secondary signal: where expert judgment and Self-RAG disagree, expert judgment takes precedence, and systematic disagreement patterns are analyzed to identify calibration biases in the technical proxy.

We compare the GIO classification against two baselines of increasing strength:

- **Keyword-Matching Baseline:** Triggers retrieval if prompts contain temporal markers (“current”, “latest”, “today”), volatility indicators (“price”, “weather”, “score”), interrogative markers (“Who”, “When”), or named entities not in a common-knowledge list.
- **Trained Classifier Baseline:** A fine-tuned BERT-based intent classifier trained on the same feature dimensions as GIO (I_{gap} , T_{decay} , E_{spec} , $V_{volatility}$) but without the structural commitments of the framework (i.e., without the mode hierarchy or the gating role of I_{gap}). This baseline tests

whether GIO’s theoretical structure adds predictive value beyond the raw features it identifies.

- **GIO Model:** Triggers retrieval based on the GIO Matrix classification (Table 1).

We evaluate all three models against both ground truth sources using paired McNemar’s tests on the classification disagreements, with the $F1$ -score as primary metric. A power analysis (G*Power, McNemar’s test for paired proportions) indicates that a sample size of $N = 800$ is sufficient to detect a medium effect size ($d = 0.3$) with $\alpha = 0.05$ and power $1 - \beta = 0.8$.

We hypothesize (**H2**) that the GIO model will achieve a significantly higher $F1$ -score than the Keyword-Matching Baseline on both ground truth measures, as determined by McNemar’s test ($p < 0.05$). We further hypothesize (**H3**) that GIO will match or exceed the Trained Classifier Baseline, demonstrating that the framework’s structural commitments (mode hierarchy, I_{gap} gating) add interpretive value without sacrificing predictive accuracy. Should the Trained Classifier significantly outperform GIO, this would indicate that the framework’s structural constraints are overly rigid—a falsifiable and informative outcome.

7 Discussion and Limitations

7.1 The Personalization Wall

GIO currently models “Public Intent.” However, RAG systems increasingly utilize *Personalized Context* (e.g., querying the user’s email or calendar). A query like “When is my flight?” has maximal GN, but zero GEO opportunity for public publishers. This “Personalization Wall” represents a fundamental boundary condition: GIO’s GEO strategies apply only to the *public retrieval* path, not to private-agent contexts.

The boundary is not static. As AI assistants deepen integration with private data stores—Gemini Live with Google Workspace, Copilot with Microsoft Graph, Apple Intelligence with on-device data—the share of queries resolved through *private* retrieval is likely to grow. Queries that today trigger public search (“What’s on my schedule tomorrow?”, “Summarize the report my team shared”) will increasingly be served from personal context, compressing the addressable space for public GEO. GIO must therefore be understood as a framework whose *strategic surface* may narrow over time, even as its *theoretical validity* (the epistemic classification itself) remains intact. Future iterations should model the Public/Private retrieval split explicitly, potentially introducing a prior probability P_{public} that gates the GEO opportunity assessment.

7.2 Alignment and Sycophancy

Optimizing for “Grounding Necessity” carries the risk of *sycophancy*—where models prioritize optimized content that aligns with user biases over objectively higher-quality but unoptimized sources. Future research must address how GEO interacts with model alignment and safety guardrails.

7.3 Epistemic Lock-in

If engines exclusively retrieve from optimized High-GN sources, a feedback loop emerges where the “truth” becomes limited to sources that structure data for machine readability. This risks marginalizing

high-quality content that is not GEO-optimized, collapsing the diversity of information available to generative systems.

7.4 Validity of the GN Hypothesis

The proposed framework model in Equation (1) is intentionally underspecified. While the interaction term ($E_{spec} \times V_{volatility}$) addresses the Specificity Paradox, other interaction effects may exist (e.g., between T_{decay} and I_{gap}). Additionally, the model assumes that GN is primarily a property of the prompt; in practice, GN may vary across model architectures with different training data distributions.

7.5 Single-Turn Scope

GIO currently models individual prompts in isolation. In practice, Grounding Necessity frequently emerges *during* a conversation rather than being fully determined by the initial query. A user may begin with a parametrically answerable question (“What is quantitative easing?”) and follow up with a high-GN refinement (“How did the ECB apply it in Q3 2025?”). This multi-turn dynamic—where GN escalates through conversational context accumulation—is not captured by the current framework. Extending GIO to model GN as a function of the full conversational state, potentially drawing on recent work in conversational search intent [19], is a necessary next step.

7.6 Multi-Domain Queries

Queries such as “Plan a trip to Japan” (Mode 3.2) trigger retrieval across heterogeneous domains (flights, hotels, cultural sites, weather, visa requirements), each with distinct volatility profiles and source types. The current framework treats GN as a scalar property of the prompt. For multi-domain queries, GN may be more accurately modeled as a *vector*—one component per domain—with the overall GEO strategy determined by the composition of domain-specific GN values. This extension is left for future work.

8 Conclusion

GIO provides the missing strategic link between the economic potential of GEO and the engineering reality of Agentic AI. By formalizing *Grounding Necessity* as an epistemic property of the prompt—and accounting for the interaction between entity specificity and volatility—GIO moves beyond descriptive taxonomies to a predictive framework.

Ultimately, if validated, GIO reframes the discipline of GEO: shifting it from mere content optimization (supplying answers) to demand shaping (understanding which questions necessitate those answers).

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